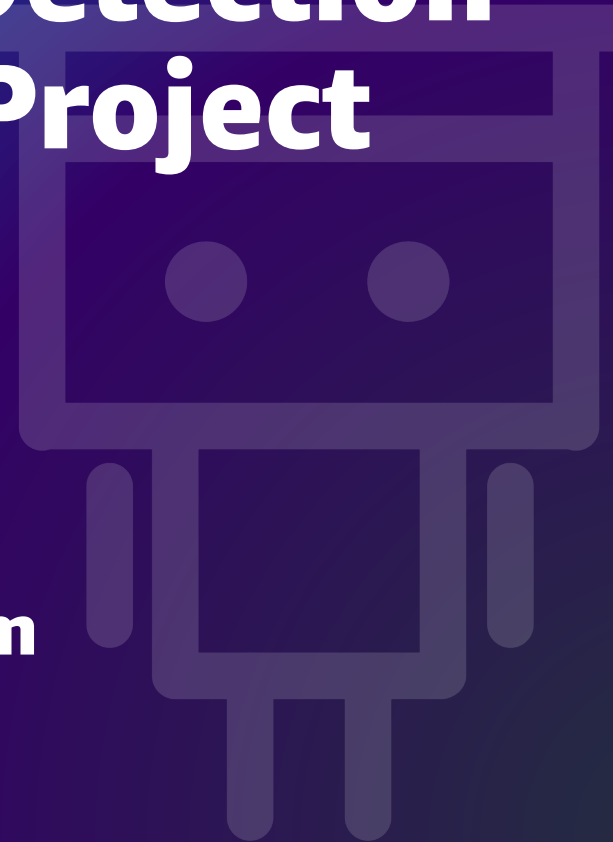


Fraudulent Claim Detection Machine Learning Project

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Agenda

- ⚙ Problem Statement
- ⚙ Data & Approach
- ⚙ Modeling Results
- ⚙ Key Insights & Feature Importance
- ⚙ Business Implications & Recommendations
- ⚙ Q&A

Problem Statement

Challenge: Manual review of insurance claims is slow and costly.

Goal: Automate fraud detection to prioritize high-risk claims, reduce losses, and improve investigator efficiency.

Scope: Use historical claim data and ML models to flag potentially fraudulent behavior.

Data & Preprocessing

- Dataset: ~1,200 claims; 30% labeled fraudulent.
- Key fields: Policyholder demographics, incident details, vehicle info, fraud flag.

- Steps:

- Impute missing values (median for numeric, mode for categorical)
- One-hot encode categoricals, scale numerics
- Oversample minority class to 1:1 balance

Insurance Claims				
	months_as_custome	age	policy_number	policy_bind_da
1	328	48	521585	2014-10-17
2	228	42	342868	2006-06-27
3	134	29	687698	2000-09-06
4	256	41	227811	1990-05-25
5	228	44	367455	2014-06-06

Feature Engineering & Selection

Initial features: ~50 mixed types

Technique:

- L1-penalized logistic regression (SAGA)
- SelectFromModel (mean-threshold)
- ⚙ **Result:** ~20 top predictors fed into downstream models.

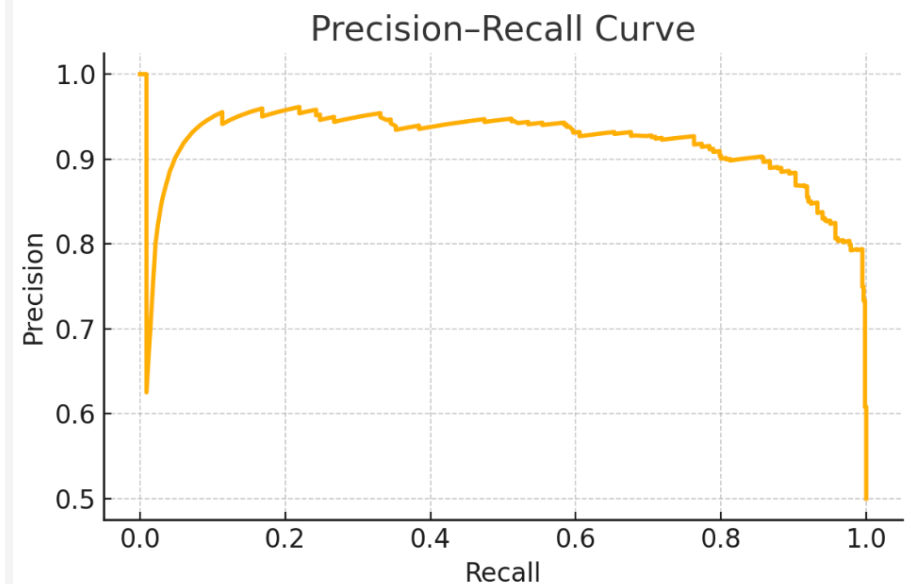
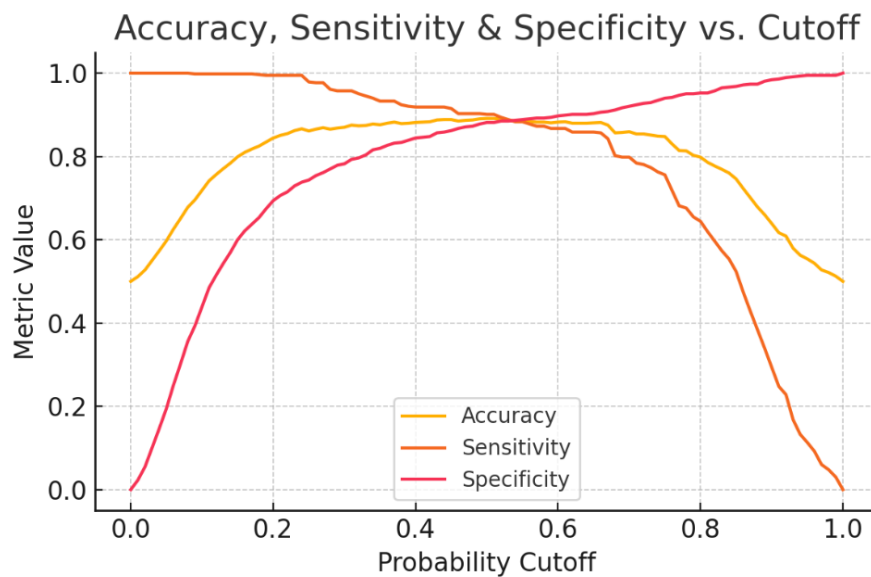
Modeling Approach

- ⚙ **Baseline:** Logistic Regression
- ⚙ Optimized decision cutoff via accuracy/sensitivity/specificity curve
- ⚙ **Advanced:** Random Forest
- ⚙ Baseline RF → extract importances → retrain on top ~12 features
- ⚙ 5-fold GridSearchCV over n_estimators, max_depth, min_samples_split/leaf, max_features

Logistic Regression Results

- ⚙ Optimal probability cutoff: ≈ 0.45
- ⚙ Training:
 - ⚙ Accuracy 0.85, Recall 0.82, Precision 0.80, F1 0.81
- ⚙ Validation:
 - ⚙ Accuracy 0.83, Recall 0.78, Precision 0.79

Logistic Regression Results



Random Forest Results

- ⚙ Best params:

- ⚙ `n_estimators=200, max_depth=20, min_samples_split=5, min_samples_leaf=2, max_features='sqrt'`

- ⚙ Training:

- ⚙ Accuracy 0.90, Recall 0.88, Precision 0.89, F1 0.89

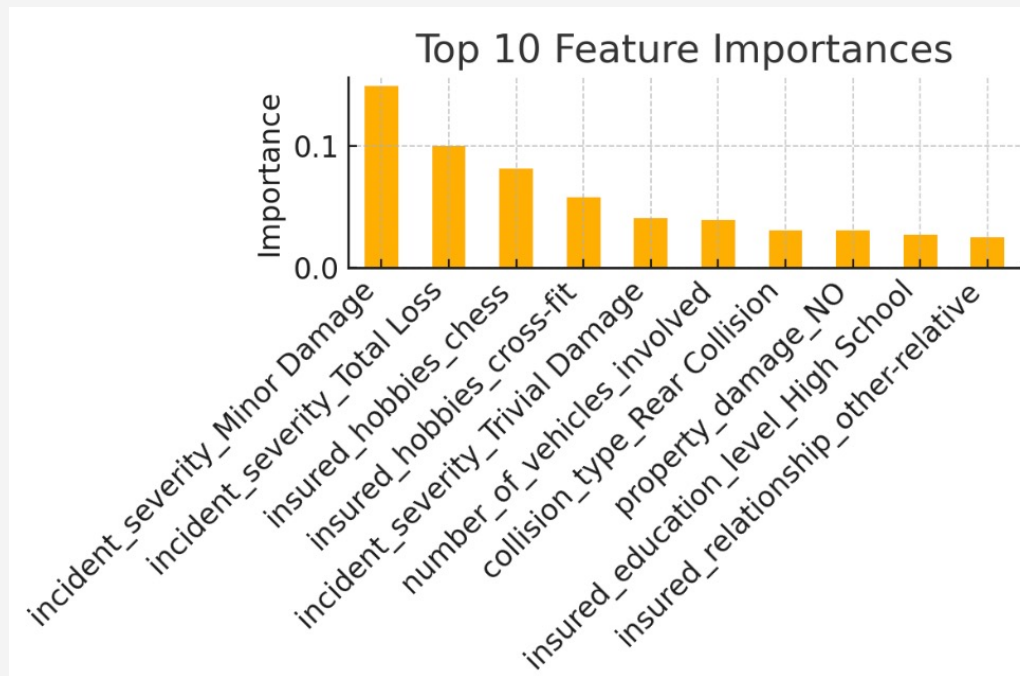
- ⚙ CV:

- ⚙ Mean train 0.90 vs. val 0.87 → minimal overfitting

- ⚙ Validation:

- ⚙ Accuracy 0.88, Recall 0.85, Precision 0.87

Random Forest Results



Feature Importance

- ⚙️ Top 5 predictors:

- » collision_type
- » property_damage
- » incident_severity
- » witness_present
- » vehicle_age

- ⚙️ **Insight:** Incident characteristics (type/severity/damage) drive most of the fraud signal.

Answering the Key Questions

- ⚙ **How detect patterns?**
 - » Preprocess historical claims, balance classes, and use feature-selection to highlight fraud-signal features.
- ⚙ **Most predictive features?**
 - » collision_type, property_damage, incident_severity, witness info, vehicle_age.
- ⚙ **Can we predict incoming claims?**
 - » Yes: RF model yields ~0.85–0.88 recall/precision on hold-out data.
- ⚙ **What insights help improve process?**
 - » Focus investigator review on claims with high-risk incident profiles.
 - » Dynamic cutoff adjustments based on operational costs.

Business Implications & Recommendations

- ⚙ **Efficiency:** Automate triage to review ~30% of claims flagged high-risk instead of 100%.
- ⚙ **Savings:** Early identification of fraud reduces payouts and investigative backlog.
- ⚙ **Scalability:** Model can retrain quarterly with fresh labels to adapt to new fraud schemes.
- ⚙ **Recommendations:**
 - » Deploy RF model as scoring service in claims pipeline (threshold ≈ 0.45).
 - » Monitor metrics (precision & recall) monthly; adjust cutoff as needed.
 - » Extend features with NLP on claim descriptions and external data (weather, crime rates).
 - » Governance: Periodic model audits to detect drift and maintain fairness.